P0 CCD Sniffer Carrier — PRELIMINARY

SYSTEM FLOW

J1962 male pigtail -> protected CCD receive daughterboard -> ESP32-S3-DevKitC-1 -> USB-C / local Wi-Fi AP

CARRIER PCB CONNECTIONS

J1 is a 2x8 internal harness header: pad number equals J1962 contact number. Only contacts 3 and 11 are tentatively routed as CCD+ and CCD-.

J2 receiver mezzanine: 1 CCD+ RAW, 2 CCD- RAW, 3 logic GND, 4 3V3, 5 RX_LOGIC, 6 RESERVED. J2 must include validated input protection and threshold/hysteresis.

J3/J4 are 1x22 sockets for the ESP32-S3-DevKitC-1 v1.1. J3-1 supplies 3V3, J3-4 is GPIO4/RX_LOGIC, J3-22 is GND; J4-1 is GND.

OPEN ITEMS — NOT FOR FABRICATION OR VEHICLE CONNECTION

Confirm Prowler DLC pin population. No pin-16 power input/regulator is implemented. Pin 5 signal ground is isolated from logic ground by default; JP1 is an optional open link.

CCD front-end protection, loading, comparator reference, hysteresis and bus waveform are not validated. P0 remains receive-only; no bias, termination or transmitter is fitted.

Power the DevKitC through its USB-C connector for this layout study. Define and validate protected vehicle power before relying on standalone operation in the car.

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